Surrogate Substitution Preserves PHI Detectability: A Multi-Detector Equivalence Study

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Abstract

Structure-preserving de-identification replaces protected health information (PHI) with realistic same-type *surrogates*—"Anna S." becomes "Maria S.", not `[NAME]`—so that clinical text stays fluent and downstream tools keep working. But this only helps if the substitution does not itself corrupt the signal those tools rely on. We ask a narrow, testable question: *on the spans a de-identifier actually masks, can downstream PHI detectors still find the surrogate?* We introduce a paired, multi-detector evaluation protocol that (i) scores utility *only on masked spans*, decoupling *coverage* from *utility*; (ii) uses *equivalence testing* (TOST) rather than null-hypothesis significance testing, which is uninformative at our sample size (56k paired spans); and (iii) builds a surrogate-failure typology separating fixable generator defects from intrinsic detector limits. Across 11 detectors, 7 benchmarks, and 7 languages (1,750 documents), recall on masked spans moves from 76.2% to 75.0%—a change our equivalence test shows is *statistically indistinguishable from zero within a* ± 2*-point margin* ($p \approx 1 \times 10^{-8}$), with detector ranking preserved. The residual loss is not detectors getting worse at PHI: it concentrates in *malformed and out-of-distribution* surrogates (truncation `Chicago` → `Illino`, salience loss `Cedars-Sinai` → `Vidant`). A redaction floor and an open-source surrogate baseline confirm the effect is a property of well-formed substitution generally, not one tool. We release the evaluation subsets and scoring code so the protocol can audit any structure-preserving transform.

1 Introduction

De-identification of clinical text has two families of output. **Redaction** deletes or tags PHI (`John Smith` → `[NAME]`), which is safe but destroys the fluency, layout, and distributional properties that downstream models and human readers depend on. **Structure-preserving** de-identification instead substitutes each PHI value with a plausible, same-type *surrogate* (`John Smith` → **`Maria Lopez`**), keeping the document readable and machine-parseable [4]. The second family is increasingly attractive: it lets de-identified data flow into analytics, model training, and even second-pass detection without breaking pipelines built for real text.

The promise of structure preservation rests on an unstated assumption: *the surrogate carries the same downstream signal as the value it replaced.* If substitution silently degrades the very features a PHI detector, an NER model, or a clinical parser relies on, then "structure-preserving" is a misnomer—the transform would be quietly laundering PHI into forms tools can no longer see or handle, which is both a utility problem and, for a second-pass safety net, a privacy problem.

This assumption is rarely tested directly, and testing it well is harder than it looks. Three pitfalls recur:

1. **Coverage confounds utility.** A de-identifier that masks 60% of PHI and one that masks 95% cannot be compared on whole-document F1—the score conflates *how much* it masks with *whether what it masks stays usable*. The two must be measured separately.

2. **The large-N significance trap.** With tens of thousands of paired spans, any non-zero difference is "statistically significant" under a standard test, even when operationally meaningless. A p-value here answers the wrong question.

3. **Aggregate error hides mechanism.** A single "-1.2 points" says nothing about *why* the loss happens—boundary jitter, detector weakness, or

a defect in surrogate generation. Only the last is fixable, and only an error typology tells them apart.

We address all three. Our contributions:

- A **paired multi-detector protocol** scoring utility on masked spans only, decoupling coverage from utility, across 11 detectors $\times$ 7 benchmarks $\times$ 7 languages (§4–§5).
- An **equivalence-testing analysis** (TOST) replacing the uninformative significance test with a bounded-effect claim: the change in detectability is provably within ± 2 points of zero (§6).
- A **surrogate-failure typology** attributing the small residual to surrogate-generation quality rather than detectors getting worse at PHI (§7).
- A set of **comparison experiments**—a redaction upper-bound and an open-source surrogate baseline—that make the result reproducible and isolate what is generic to *any* substitution versus specific to one tool (§5, §6).

The framing is deliberately not "a proprietary tool is good." It is: *here is how to measure whether a structure-preserving transform preserves utility, rigorously*, with one commercial transform as the case study and open baselines for reproducibility.

2 Related Work

2.1 De-identification and surrogate substitution

Clinical de-identification is a long-studied sequence-labeling problem [31], anchored by the i2b2/n2c2 and MEDDOCAN shared tasks [23, 29] and by systems ranging from rule- and dictionary-based pipelines [25] through recurrent and transformer taggers [7, 16, 20] to instruction-tuned LLMs used zero-shot [21]. Most of this literature optimizes *detection*, typically against the HIPAA Safe Harbor identifier set [30]. The downstream question—*what to put in the PHI's place*—is comparatively under-studied. Surrogate ("hiding in plain sight") replacement was proposed to keep de-identified notes realistic and to resist re-identification [4]; subsequent work showed poor surrogates can even aid re-identification (the "parrot" attack), underscoring that surrogate *quality* matters [5]. Our work is orthogonal to the detection literature and to any particular surrogate generator: given that a span is masked, we ask whether the *replacement* remains detectable and well-formed.

[Citation pending] The commercial transform used as our case study implements a surrogate-generation method covered by a Custodian Labs patent [11]. The formal citation will be finalized once the patent issues; we cite it as the source of the transform rather than describing or extending the method.

2.2 Utility-preservation evaluation

Whether privacy transformations preserve downstream utility is central to privacy-preserving NLP. Prior evaluations typically run a single downstream model on original vs. transformed data and compare end-task scores. Two weaknesses recur: (a) a single downstream model cannot separate "the transform is fine" from "this model is robust," and (b) whole-corpus metrics mix masked and unmasked content. We address (a) with an 11-detector panel spanning rule-based, fine-tuned, and LLM detectors, and (b) by restricting utility measurement to masked spans.

2.3 The large-N trap and equivalence testing

When samples are large, null-hypothesis significance testing rejects the null for negligible effects; the p-value measures precision, not importance—a hazard increasingly flagged in NLP evaluation [3, 9]. The standard remedy is *equivalence testing*—the two one-sided tests (TOST) procedure [2, 28]—which specifies an equivalence margin Δ and tests whether the effect is provably inside $[-\Delta, +\Delta]$; see Lakens [17], Lakens et al. [18] for practical tutorials. TOST is standard in biostatistics but under-used in NLP evaluation, where large paired corpora make the trap acute. We adopt it as the primary inferential tool and report McNemar's paired test [24] only to demonstrate the trap.

3 Problem Formulation

Structure-preserving de-identification. A transform T maps a document d to d' by replacing each detected PHI value v (of type τ) with a surrogate s of the same type, leaving all other characters unchanged. Gold PHI spans on d are re-projected onto d' by character-level alignment, giving paired spans (v, s).

Coverage vs. utility. Two quantities must not be conflated. *Coverage* is the fraction of true PHI that T detects and replaces (a property of T's detector). *Utility* is, given that a span was replaced, whether a downstream detector still finds the surrogate s as well as it found v (a property of T's generator and of the surrogate's realism). We report coverage separately and measure utility *only on the masked-span population* $\{(v, s)\}$. This prevents a low-coverage transform from looking good—or a high-coverage one from looking bad—on a metric that is really about substitution quality.

4 Evaluation Protocol

Figure 1 summarizes the protocol. **Paired multi-detector design.** Each document is scored in two conditions—original d and transformed d'—by the *identical* detector suite D ($|D| = 11$). Because the only change between conditions is the substitution, any per-span change in detection is attributable to the substitution, not to the detector or the document.

Metrics. We report span-level P/R/F1 under three matching modes—*exact* (start+end+type), *type* (type+boundary), *overlap* (any character overlap+type)—and *leakage* $= 1 -$ recall, the HIPAA-critical quantity. The headline utility metric is *recall retention on masked spans* $=$ recall(d')/recall(d) restricted to $\{(v, s)\}$, under overlap matching (so pure boundary jitter from length changes, "Anna S."→"Maria S.", is not charged as a miss).

Equivalence testing. For pooled masked-span recall we run TOST with margin $\Delta = 2$ points: we reject non-equivalence iff the 90% CI of the recall difference lies entirely within $[-2, +2]$. We also report McNemar's test to illustrate the large-N trap. We sweep $\Delta \in \{1, 2, 3\}$.

Error attribution. For the lost population (found on d, missed on d') we hand-code each span into a small failure typology (§7) and check length-preservation to separate boundary artifacts from genuine misses.

5 Experimental Design

Detector panel D. Eleven detectors, three families: *rule/statistical*—Microsoft Presidio [25] (built on spaCy [15]), OBI `deid_roberta` [16], a RoBERTa tagger [8, 19]; *open LLMs (local)*—Gemma 4 31B / E4B [13], Qwen 3.5-{4B, 9B, 35B-A3B} [27], Llama 3.1-8B / 3.3-70B [14], DeepSeek V2-Lite [6]; *frontier API*—OpenAI GPT-5 [26]. All are Transformer-based [32]. The panel is deliberately heterogeneous: if the equivalence result held only for one architecture it would be a model artifact, not a property of the transform. (Model-version citations point to the closest published technical report for each family. One further model, Moonshot Kimi-VL-A3B, was attempted but excluded because it required non-standard model-loading code.) LLM detectors emit free-text or JSON identifier lists; we recover character spans by exact-then-fuzzy string search over the source document, so all detectors are scored on the same span basis.

Benchmarks (7; 250 docs each, 1,750 total). ASQ-PHI (English clinical queries), MEDDOCAN (Spanish clinical) [23], MultiCoNER v2 (multilingual NER) [12, 22], and PII-Masking-300k [1] in English, Dutch, French, German. Together: 7 languages, clinical and general-PII text, free-form and structured (JSON) formats. The transform under test is Custodian Guardian Layer `transform` mode (top-1 surrogate, `pii_entities=ALL`).

Comparison conditions. The core study answers "does *this* transform preserve detectability." Three contrasts isolate *why* and *how generally*:

- **C1—Redaction upper-bound.** Replace each masked value with $\star\star\star\star\star$ (no surrogate signal) and re-detect; the transform−redact gap quantifies what structure preservation buys.
- **C2—Open-source surrogate baseline.** Replace the commercial generator with an open one (Faker [10]) on the *same* masked spans; tests whether the result is generic to substitution or tool-specific, and makes the pipeline reproducible.
- **C3—Per-benchmark equivalence.** Run TOST within each benchmark, sweeping Δ, to check the pooled claim is not an averaging artifact.

The core paired study, the equivalence analysis (pooled and per-benchmark), the redaction floor, the open-surrogate baseline, and the error typology are complete; C1/C2 are currently demonstrated with the CPU detector (Presidio), with a full-panel extension noted as the next strengthening step.

6 Results

Detectability is statistically unchanged. Restricting to the 56,174 masked spans and pooling all 11 detectors, recall moves 76.2% → 75.0% (-1.2 pts; 95% CI $[-1.5, -1.0]$). The TOST equivalence test ($\Delta = 2$) rejects non-equivalence at $p \approx 1 \times 10^{-8}$: the change is provably bounded within ± 2 points of zero. Ranking is preserved; the flagship Llama 3.3-70B is essentially unchanged (ΔF1 $+0.003$). The same contingency is

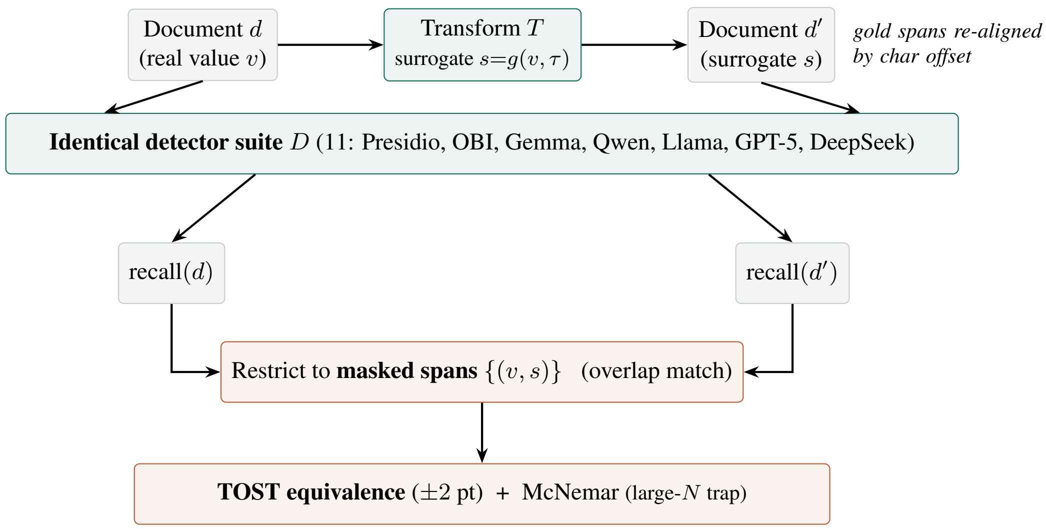

Figure 1: The paired evaluation protocol. Each document is scored in *both* conditions (d, d') by the identical 11-detector suite, so any per-span change is attributable to the substitution. Utility is measured *only on masked spans*, and the paired recall difference is assessed with an equivalence test (TOST) rather than a significance test (§4).

"significant" under McNemar's test—a direct demonstration of the large-N trap.

Whole-document view (conservative). Before restricting to masked spans, Table 1 reports span-level F1 and leakage on *all* gold spans, per detector. Whole-document ΔF1 ranges from $+0.003$ (Llama 3.3-70B) to -0.047 (Qwen 3.5-35B-A3B), and leakage rises by only $+0.3$ to $+4.1$ points. This view is deliberately pessimistic—it mixes masked spans with exact-boundary penalties on length-changed surrogates and with spans the transform never touched—yet it already bounds the worst case (no detector's mean F1 moves by more than 4.7 points) and preserves ranking across a 100$\times$ span of detector quality (F1 $0.76 \rightarrow 0.04$). The masked-span analysis below isolates the substitution effect from this dilution.

Recall retention on masked spans (overlap). Restricting to masked spans removes that dilution. When the transform masks a span, detectors still find the surrogate 93–100% of the time (Table 2); the $\sim$3-point exact-boundary drop is a length-jitter artifact that vanishes under overlap matching. (The two floor detectors, Llama 3.1-8B and OBI `deid_roberta`, are omitted from Table 2: their original recall is so low that the retention ratio is dominated by noise.)

Per-benchmark equivalence (C3). Table 3 (57,112 masked spans, via `scripts/analyze_equivalence.py`) shows

System	F1		Δ	Leak	
	orig	transf	F1	orig	transf
Gemma 4 31B	.755	.710	$-$.045	.255	.285
Gemma 4 E4B	.737	.698	$-$.038	.276	.311
Llama 3.3-70B	.725	.728	+.003	.245	.262
Qwen 3.5-35B-A3B	.715	.668	$-$.047	.254	.295
OpenAI GPT-5	.705	.674	$-$.032	.308	.333
Qwen 3.5-9B	.655	.621	$-$.034	.391	.422
Qwen 3.5-4B	.567	.533	$-$.034	.483	.515
Presidio	.416	.398	$-$.018	.553	.570
DeepSeek V2-Lite	.409	.384	$-$.025	.672	.696
Llama 3.1-8B	.391	.376	$-$.015	.633	.650
OBI `deid_roberta`	.041	.040	$-$.002	.941	.944

Table 1: Whole-document view (all 11 detectors), mean over 7 benchmarks. Span-level F1 (type match) and leakage (1$-$recall). Conservative: it includes untouched spans and boundary penalties. Ranking is preserved and ΔF1 is bounded.

the pooled equivalence is *not uniform*: four benchmarks are equivalent within ± 2 points; MEDDOCAN and PII-nl require ± 3 (both dense, identifier-heavy, non-English—where surrogate generation is hardest, §7); MultiCoNER has too few masked spans (220) to test. The honest claim: *detectability is equivalent within ± 2 points on average and on clean text, and within ± 3 on the hardest identifier-dense text*—the residual is

System	Exact	Overlap
Gemma 4 31B	93.1	**99.8**
Qwen 3.5-9B	92.0	**100.0**
Qwen 3.5-35B-A3B	90.0	**98.4**
Llama 3.3-70B	91.4	**98.3**
GPT-5	91.9	**98.3**
Gemma 4 E4B	88.9	**98.2**
Qwen 3.5-4B	84.7	**98.0**
Presidio	91.3	**97.4**
DeepSeek V2-Lite	87.7	**92.8**

Table 2: Recall retention on masked spans (%), transformed÷original.

Benchmark	Masked	O→T	McN. χ^2	Margin
ASQ-PHI	5,427	91.9→91.1	8.0	±2 ✓
MEDDOCAN	31,978	73.4→71.5	92.4	±3 ✓
MultiCoNER	220	71.8→71.4	0.0	underpow.
PII en	5,181	76.0→75.1	4.0	±2 ✓
PII nl	3,757	78.2→76.3	13.1	±3 ✓
PII fr	5,984	75.2→75.9	3.5	±2 ✓
PII de	4,565	76.3→76.5	0.4	±2 ✓
Pooled	**57,112**	**76.1→74.9**	**82.1**	±2 ✓

Table 3: Per-benchmark equivalence. O→T = masked-span recall, original→transform (%). Pooled TOST Δ=2: p=3×10^{-9}.

concentrated and attributable, not diffuse degradation.

Redaction floor (C1). Replacing masked values with $\star\star\star\star\star$ and re-detecting with Presidio (Table 4): transform recall tracks the *original* within a few points, while redaction collapses to ≈0–4% (the residual is spurious overlap on the $\star$ run). Structure-preserving substitution preserves essentially the entire detectability that redaction destroys.

Open-surrogate baseline (C2). Replacing the commercial generator with Faker on the same masked spans (Table 5) yields two findings. *Generality*: an open substitutor preserves masked-span recall at least as well as the original—detectability preservation is a property of well-formed same-type substitution, not one vendor, and reproduces without proprietary access. *The residual is generator quality*: Faker, emitting clean canonical values, *exceeds* the commercial transform by 18–20 points on the hard non-English benchmarks, precisely where the commercial surrogates truncate or garble (§7).

Leakage barely moves (+1.7 to +4.1 pts across

Benchmark	Masked	Orig.	Transf.	Redact
ASQ-PHI	531	94.5	95.7	**0.0**
MEDDOCAN	2,925	75.4	72.9	**0.0**
PII en	471	75.6	73.5	**1.9**
PII nl	373	83.6	77.7	**3.8**
PII fr	544	61.4	59.9	**2.2**
PII de	415	56.1	58.3	**0.0**

Table 4: C1 redaction floor: Presidio masked-span recall (%).

Benchmark	Masked	Orig.	Custodian	Faker
ASQ-PHI	531	94.5	95.7	**97.7**
MEDDOCAN	2,925	75.4	72.9	**79.7**
PII en	471	75.6	73.5	**89.2**
PII nl	373	83.6	77.7	**93.0**
PII fr	544	61.4	59.9	**78.5**
PII de	415	56.1	58.3	**78.1**

Table 5: C2 open-surrogate baseline: Presidio masked-span recall (%). Detector-independent argument; Presidio-only demonstration.

systems), so surrogates are not systematically easier to miss than the PHI they replace.

Where the loss lands (per benchmark). Table 6 breaks whole-document ΔF1 out by benchmark and detector. The pattern is sharp: almost all of the loss concentrates on **ASQ-PHI**—short, sparse-PHI adversarial queries where a single substitution dominates the document score (-0.09 to -0.18)—while the other six benchmarks are essentially flat ($|\Delta\text{F1}| \leq 0.06$, most ≤ 0.03). Even the fine-tuned OBI tagger and the rule-based Presidio move by at most a few thousandths on most benchmarks. This is the first sign that the effect tracks *surrogate-generation difficulty* (short adversarial text, dense non-English identifiers) rather than detector family—made precise in the error analysis (§7).

What the transform does (examples). Table 7 shows the substitution qualitatively across languages and formats: each PHI value becomes a same-type surrogate while clinical shorthand, foreign-language syntax, and JSON structure are left byte-for-byte intact. On the well-formed English cases every surrogate is still detected.

Coverage, reported separately. All results above are measured *on the spans the transform masks*. Coverage—how much true PHI it detects and replaces—is a distinct, detector-side axis. It tracks how closely

System	ASQ-PHI	MEDDOCAN	PII en	PII nl	PII fr	PII de	MultiCoNER
Gemma 4 31B	−.175	−.017	−.043	−.038	−.014	−.016	−.009
Gemma 4 E4B	−.106	−.060	−.032	−.021	−.018	−.025	−.007
Qwen 3.5-35B-A3B	−.132	−.034	−.026	−.034	−.040	−.054	−.007
OpenAI GPT-5	−.098	−.046	−.049	−.019	+.008	−.005	−.014
Qwen 3.5-9B	−.123	−.017	−.025	−.010	−.034	−.024	−.008
Qwen 3.5-4B	−.086	−.055	−.034	−.028	−.015	−.023	+.002
Presidio	+.003	−.037	−.032	−.015	−.020	−.015	−.007
Llama 3.1-8B	−.108	+.023	−.014	+.007	−.009	+.003	−.006
OBI `deid_roberta`	+.001	−.008	−.004	−.001	+.003	−.002	−.001

Table 6: Per-benchmark whole-document ΔF1 (transformed − original; negative = drop). Almost all loss lands on ASQ-PHI; the other six benchmarks are flat. Llama 3.3-70B and DeepSeek V2-Lite omitted for space (both flat; see Table 1 for their pooled ΔF1).

Case	Original	Transformed
ASQ-PHI (clinical query)	...a 34-year-old female ...like Anna S., ...at Methodist Hospital on April 12, 2023?	...a **35**-year-old female ...like **Maria S.**, ...at Methodist Hospital on **March 13, 2021**?
ASQ-PHI (shorthand)	Rec mgmt of 70yo M w/ CHF, seen by Dr. John L. at Mt. Sinai on Feb 21, 2023. alt tx options...	Rec mgmt of **73yo** M w/ CHF, seen by Dr. **James L.** at **Mt. Egypt** on **Nov 19, 2021**. alt tx options...
PII (German)	...ermächtigen hiermit Monsignore, ...Mit Datum 23/07/2011...	...ermächtigen hiermit **Fulgenzio**, ...Mit Datum **24/07/2011**...
PII (French, JSON)	`{"Date":"20/05/2022", "City":"Saint-Priest", "Username":"phprosdocimo"}`	`{"Date":"21/05/2023", "City":"Saint-Priest", "Username":"phprosdocimo"}`

Table 7: Worked transform examples. Only the sensitive value moves; abbreviations (`w/ CHF`, `alt tx`), non-English syntax, and JSON keys/quotes/indentation are preserved, so downstream parsers and detectors keep working. In the JSON case the transform changes only the date value and leaves the (also-sensitive-looking) username untouched—a coverage decision, not a utility one (see below).

a benchmark's annotation matches the transform's notion of sensitive content: it masks **80.9%** of gold PHI on ASQ-PHI and **48.3%** on MEDDOCAN, and less (≈26%) on general-domain NER/PII corpora whose annotated entities (encyclopedic names, generic places) fall outside that scope; a configuration sweep confirmed `domain=General` maximizes coverage. A per-entity-type breakdown and a detection-vs-replacement diagnosis (the coverage gap is mostly a *replacement*-step issue—≈77% of missed clinical identifiers were flagged by the transform's own detector but not substituted) are in Appendix A. Decoupling matters because the two failure modes have different owners and fixes: an under-masked span is a *detection/replacement* miss, whereas a masked-but-missed surrogate is a *generation* defect (§7). Conflating them on whole-document F1 (Table 1) would let a high-coverage/low-quality transform and a low-coverage/high-quality one look identical. We therefore report coverage as context and reserve the utility claim for masked spans.

Matching modes. Unless noted, retention uses *overlap* matching (detector flags any part of the surrogate span). Exact matching (Table 2, left) additionally requires identical character boundaries and is therefore sensitive to surrogate length changes ("Anna S."→"Maria S." shifts the end offset); the ∼3-point exact−overlap gap is this boundary jitter, not missed PHI. Type matching (boundary + type) sits between the two and tracks overlap closely.

7 Error Analysis

The lost population. Pooled across detectors and masked spans: 39,550 spans found in both conditions; 3,261 lost. Critically, 51% of lost spans are the *same length* as the original—not a boundary effect. By type: LOCATION 28%, NAME 23%, DATE/AGE 22%, ID/contact 20%.

Three failure modes—all generator-side. (1) *Malformed/truncated surrogates* (largest cause): `Chicago` → **`Illino`**, `El Paso` → **`El`**, `Ciudad de la Habana` → **`Cuidad de la Havana`**. The fragment no longer matches the lexical pattern detectors learned for real names/places; this is why span-level loss is highest on MEDDOCAN (6.9%; Spanish, dense, identifier-heavy) and lowest on clean English ASQ-PHI (2.6%). (2) *Loss of salience*: a canonical entity replaced by an obscure one (`Cedars-Sinai` → **`Vidant`**); detectors partly rely on pre-training familiarity, so swapping a famous value for a rare one removes the prior. Inherent to any value substitution; mainly costs weaker detectors. (3) *x-masking of IDs/emails*: `nachorutor@...` → **`nxxxxxxxxx@...`**. The `x` run preserves format but breaks the realistic-token pattern. This case is *privacy-positive*—the original value is destroyed—even though it counts against recall.

Implication. Detectors are not getting worse at PHI; the small recall gap is driven by surrogate-generation quality (truncation, garbling, salience, `x`-masking). C2 confirms this directly: an open generator emitting clean canonical values *exceeds* the commercial transform by 18–20 points on exactly the hard non-English benchmarks where its surrogates truncate—so the residual tracks generator quality, not the act of substitution.

8 Discussion

What the evidence supports. Structure-preserving substitution does not hide well-formed PHI from downstream detection. Because the effect is equivalence-bounded within ± 2 points and ranking-preserving across a heterogeneous 11-detector panel, a transform of this quality can be safely inserted ahead of detection/analytics pipelines built for real clinical text. **Generality (C2).** Faker preserves masked-span recall at least as well as the original across all six benchmarks, so ± 2 points is a property of well-formed substitution, not one vendor; where the commercial generator trails, a cleaner generator closes the gap, locating the residual squarely in generation quality. **A reusable protocol.** The paired masked-span design and the TOST margin are not specific to one vendor or language—a template for auditing any structure-preserving privacy transform.

9 Ethics and Data Statement

All benchmarks are public or synthetic (ASQ-PHI synthetic; MEDDOCAN released for a shared task; PII-Masking-300k synthetic; MultiCoNER v2 public). **No real patient data is used.** We release the 250-document subsets and scoring code with license notes. **Dual-use.** A utility-preserving de-identifier could in principle launder identifiable data into a fluent form; our masked-span/leakage reporting and the privacy-positive framing of `x`-masking keep the privacy accounting explicit. **Conflict of interest.** The commercial transform (Custodian Guardian Layer) is developed by Custodian AI, with which the authors are affiliated; this work was conducted with Custodian AI's support. To limit bias we (i) frame the contribution as a reusable protocol, (ii) include open-source baselines so results are reproducible without proprietary access, and (iii) report leakage and coverage alongside utility.

10 Limitations

Coverage is reported but not the focus; a transform can preserve utility on what it masks while under-masking (the axes are independent by design). 250 docs/benchmark bounds per-benchmark power (though the pooled masked-span N is large). C1/C2 are currently demonstrated with a CPU detector (Presidio); a full 11-detector extension is future work. LLM detectors are prompt-sensitive; we fix one prompt per model and release it with the code and data at the project's reproducibility page. Finally, all seven benchmarks are public or synthetic; validating the protocol on real-EHR corpora (n2c2, MIMIC-IV-Note, CARMEN-I) is future work pending the relevant data-use agreements.

A Coverage Details

Coverage (§6) is the fraction of a benchmark's gold PHI whose characters the transform actually changed (difflib alignment). Table 8 breaks it down by entity type on the clinical sets (ASQ-PHI + MEDDOCAN). High-frequency free-text types (names, dates) are masked well; the weak spots are high-sensitivity structured identifiers and geography—exactly the HIPAA Safe Harbor items that most need masking.

Detection vs. replacement. On a sample of

Entity type (clinical)	Coverage
NAME / person	75.0%
DATE / age	73.4%
ID / contact (MRN, patient ID, phone, email)	50.7%
ORG / facility	46.6%
LOCATION / address	43.7%

Table 8: Coverage by entity type on clinical benchmarks. Overall the transform altered 39% of annotated PII (54% on clinical sets); general-domain corpora sit near 26%.

8 MEDDOCAN documents (62 missed ID/location spans), $48/62$ ($\approx$77%) of the *un*masked identifiers were nonetheless flagged as sensitive by the transform's own detector—they were detected but not substituted; only $14/62$ (23%) were undetected. The coverage gap is therefore mostly a *replacement-step* issue (act on everything the detector surfaces), which is more tractable than raising detection recall. (Small sample; "detected" judged by loose token overlap, so 77% is directional.) Concrete unmasked HIPAA identifiers included patient IDs (`80926`, `7845693`), care-contact IDs (`4387684`), and facilities/geography (`Hospital de Cruces`, `España`, postal codes `41005`, `28047`).

B Representative Per-Detector Examples

Agreement on well-formed surrogates. On clean English (ASQ-PHI), a plausible same-type surrogate is caught by nearly the whole panel (Table 9): the substitution is invisible to detection. Divergence is confined to the two floor detectors (Llama 3.1-8B, DeepSeek V2-Lite).

Document	Surrogate span (type)	Caught
asq_00000	Maria S. (NAME)	10/11
asq_00000	Methodist Hospital (LOC)	9/11
asq_00000	March 13, 2021 (DATE)	9/11
asq_00003	James L. (NAME)	11/11
asq_00003	Mt. Egypt (LOC)	11/11
asq_00003	Nov 19, 2021 (DATE)	11/11

Table 9: Number of detectors (of 11) that find each well-formed surrogate. Same-type swaps stay broadly detectable.

Representative losses, by failure type. Table 10 shows genuine "lost" cases (MEDDOCAN): the original value was found by most detectors, but a defective surrogate is found by few. Each maps to one of the three failure types in §7, and the drop is shared across the panel—i.e. it is a property of the surrogate, not of any one detector.

Original $\to$ surrogate	O $\to$ T	Type
`Cuba` $\to$ **`Havana`**	9 $\to$ 2	salience
`San Fernando` $\to$ **`Francsico Luis`**	8 $\to$ 3	garbled
`Cádiz` $\to$ **`Cadiz`**	8 $\to$ 4	accent stripped
`Cecilio Pujazón` $\to$ **`Pujazón`**	8 $\to$ 4	truncation
`ignaciotorne@...` $\to$ **`ixxxxxxxxxxx@...`**	9 $\to$ 2	x-mask
`23/07/1948` $\to$ **`2xxxxxxxx`**	11 $\to$ 4	x-mask
`40` $\to$ **`35`** (age)	9 $\to$ 2	bare number

Table 10: Representative lost spans on MEDDOCAN. O $\to$ T = detectors finding the original vs. the surrogate (panel of 11–12; original includes one extra decoding variant). Losses are shared across detectors and align with the §7 typology.

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